Mukesh Patel School of Technology Management & Engineering

Log Book for Capstone Project

Department : Computer Engineering
Program : B.Tech in Computer Engineering
Semester : [Semester]
TITLE OF THE PROJECT : GitaGuide: An Emotion-Aware, Explainable and Safety-Gated Guidance System Grounded in the Bhagavad Gita and the Arthashastra

Name of the Faculty Mentor: **Dr. Suresh Kurumbanshi**

STUDENT DETAILS

	NAME	ROLL NO.	CONTACT
Student 1	Harikrishna Panchal	[Roll No.]	Harikrishna.Panchal505@nmims.in

Name of group members:

Roll No.	Name	Sign (Students)	Grades (By Faculty Mentor)
[Roll No.]	Harikrishna Panchal		

Week No: 1
Date of Reporting:
From 13/07/2026 to 19/07/2026
Work carried out
• Chose the project topic: a guidance system built on the Bhagavad Gita and the Arthashastra.
• Read news reports about Gita chatbots that gave harmful answers, to understand the problem.
• Wrote down what the system must do: check safety first, understand emotion, and cite real verses.
• Prepared a short summary of the idea to show my guide.
Faculty Remark
Faculty Signature with date

Week No: 2
Date of Reporting:
From 20/07/2026 to 26/07/2026
Work carried out
• Started reading research papers on scripture-based chatbots and on AI for emotional support.
• Read how language models are used in mental health and what risks the reviews report.
• Made a table of the papers I read, with what each one does and what it does not do.
• Noticed early that I could not find a system that combined emotion, two scriptures and a safety check.
Faculty Remark
Faculty Signature with date

Week No: 3
Date of Reporting:
From 27/07/2026 to 02/08/2026
Work carried out
• Read more papers on knowledge graphs, graph-based retrieval and crisis detection.
• Read papers on bias, emotional dependency and the harm caused by sudden refusals in chatbots.
• Kept adding to my comparison table of existing systems.
• Started writing the literature review.
Faculty Remark
Faculty Signature with date

Week No: 4
Date of Reporting:
From 03/08/2026 to 09/08/2026
Work carried out
• Searched again on purpose for papers that disagree with my idea, such as work on bias and dependency.
• Found five more papers that changed some of my statements.
• Finished reviewing 34 papers and wrote down the research gap.
• Wrote the five requirements that my system has to meet.
Faculty Remark
Faculty Signature with date

Week No: 5
Date of Reporting:
From 10/08/2026 to 16/08/2026
Work carried out
• Designed the order of steps for one message: emotion, intent, safety check, memory, retrieval, answer, output check.
• Decided that the safety check must run before everything else.
• Chose the tools: FastAPI, a Neo4j graph, PostgreSQL, a hosted language model and a React interface.
• Drew the first block diagram of the system.
Faculty Remark
Faculty Signature with date

Week No: 6
Date of Reporting:
From 17/08/2026 to 23/08/2026
Work carried out
• Set up the project folders, the code layout and the testing setup.
• Looked for free and legal sources for the text of the Gita and the Arthashastra.
• Planned how to label verses by emotion and theme and how to connect them in a graph.
• Planned how I would test the system: safety prompts and a set of test queries with correct verses.
Faculty Remark
Faculty Signature with date

Week No: 7
Date of Reporting:
From 24/08/2026 to 30/08/2026
Work carried out
• Made the first working version that answers with cited verses from start to end (24 Aug).
• Added the Arthashastra (3,135 sutras) and rebuilt the knowledge graph with both texts.
• Left out very common labels when connecting verses, so the graph stays small enough for the free tier.
• Changed the Gita translation to Purohit's, because it is public domain.
• Wrote 40 test queries with the correct verses marked, and recorded the first result.
Faculty Remark
Faculty Signature with date

Week No: 8
Date of Reporting:
From 31/08/2026 to 06/09/2026
Work carried out
• Checked the live system with the database and the graph, and wrote down what I found (2 Sep).
• Made the crisis check stronger and added a backup language model (5 Sep).
• Built the first chat screen with a consent page and a crisis panel.
• Made the system run without PyTorch and measured how long each step takes.
• Added rate limits, consent saved on the server, feedback buttons and a time limit for answers.
• Prepared the deployment files, a smoke test and the first notes for testers (6 Sep).
Faculty Remark
Faculty Signature with date

Week No: 9
Date of Reporting:
From 07/09/2026 to 13/09/2026
Work carried out
• Made the embedding model smaller (INT8) so the system fits in 512 MiB of memory; the peak was 429 MiB.
• Made the system handle one question at a time to keep memory low.
• Added a check that catches broken answers, such as repeated words.
• Made replies match the user’s language: English, Hindi or Hinglish.
• Changed answers to be short and conversational, with the verse reference shown beside each verse.
• Added a router so that a greeting does not get a verse (13 Sep).
Faculty Remark
Faculty Signature with date

Week No: 10
Date of Reporting:
From 14/09/2026 to 20/09/2026
Work carried out
• Made the router faster by handling common phrases without calling the model.
• Redesigned the screen with a sidebar, regenerate and edit buttons, and optional sign-in.
• Fixed a deployment bug where the data folder was missing from the image.
• Moved the API to an AWS EC2 server in Mumbai; one turn became about 12 seconds faster.
• Added a keep-alive so the free graph database does not pause.
• Fixed a bug where chats stayed visible after signing out (20 Sep).
Faculty Remark
Faculty Signature with date

Week No: 11
Date of Reporting:
From 21/09/2026 to 27/09/2026
Work carried out
• Went through all the measurements so far and listed the weak points: retrieval, intent and faithfulness.
• Decided which extra tests I needed before writing the report.
• Planned a larger test set, a human rating study and a safety red-team set.
• Cleaned up the code and my notes.
Faculty Remark
Faculty Signature with date

Week No: 12
Date of Reporting:
From 28/09/2026 to 04/10/2026
Work carried out
• Wrote the tools to compare retrieval variants fairly: keyword only, vector only and with the graph.
• Wrote a guide and tools so that two people can judge the retrieved verses.
• Wrote 33 red-team prompts, including harmful questions framed as duty, and ran them on the system.
• Ran an automatic faithfulness check on 69 answers; one judge model failed, so I used the one that worked.
• Wrote my literature review paper and a cover letter (30 Sep).
Faculty Remark
Faculty Signature with date

Week No: 13
Date of Reporting:
From 05/10/2026 to 11/10/2026
Work carried out
• Judged the retrieved verses for 100 queries together with a second person.
• Rated 30 answers with the second person on relevance, faithfulness, helpfulness and appropriateness.
• Started writing the research paper.
• Wrote the first draft of the interim report.
Faculty Remark
Faculty Signature with date

Week No: 14
Date of Reporting:
From 12/10/2026 to 18/10/2026
Work carried out
• Re-ran the retrieval tests, turning one part off at a time, and compared the results with a statistical test.
• Tested the emotion model on 1,500 public GoEmotions examples (macro-F1 0.451).
• Sent 100 test queries to the live system to check intent classification (65 of 100 correct).
• Re-ran the 33 red-team prompts on the live system and found 2 answers that wrongly allowed violence.
• Added a fifth output rule to block such answers and 4 new tests (tests passing went from 907 to 911).
• Finished the interim report (17 Oct).
Faculty Remark
Faculty Signature with date

Week No: 15
Date of Reporting:
From 19/10/2026 to 25/10/2026
Work carried out
• Revised the research paper: fixed tables, numbers and wording after review comments.
• Made the emotion table list all 28 labels and saved the full results in a file.
• Wrote the Black Book following the college template (55 pages).
• Made the language of the reports simpler and plainer.
• Prepared this log book.
Faculty Remark
Faculty Signature with date